PHYSICAL CHEMISTRY CUMULATIVE EXAM

November 12, 1994

1. (A) A mercury lamp emits ultraviolet light at 313 nm wavelegth. If the light output is 10 W, how many photons are emitted per second? Light at this wavelength is strongly absorbed by aliphatic ketones which are decomposed by it. The quantum yield of the decomposition reaction is about 0.21 for 4-heptanone. How long must a 10 g sample of 4-heptanone be irradiated by the above lamp to decompose half of the material, assuming that all light from the lamp is absorbed?

(B) The work function (the energy needed to knock an electron free from a metal) for titanium is 4.33 eV. Calculate the kinetic energy and the velocity of the electrons ejected from Ti surface by light of 211 nm wavelength.

(C) Calculate the wavelength of an electron accelerated by a potential difference of (a) 10 V and (b) 10 kV. Which one would be appropriate to determine interatomic distances using electron diffraction?

(D) Estimate wavelengths of X-rays used in medical X-ray equipment.

Constants: Planck's constant h=6.626E-34 Js
1 eV = 1.602E-19 J
electron mass m_e = 9.110E-31 kg
speed of light c=2.998E8 m/s
Avogadro's number N_A = 6.022E23 molecules/mol